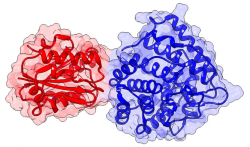
Protein Secondary Structure Classification for PPI Site Prediction using ASPRA distace

Michela Quadrini
School of Science and Technology, Camerino (MC), Italy

Carlo Ferrari
Department of Information Engineering, University of Padova, Italy

Background. Proteins consist of one or more sequences of amino acids that fold back on themselves by determining **shapes**.

The configuration is strictly related to its **function** and influence the interaction with other molecules [1]. The interactions between two proteins are referred to as **protein-protein interactions** (PPIs).



The dysfunction or malfunction of PPI cause several diseases, among which neurodegenerative disorders and cancer.

The prediction of interactions sites helps to

- understand how a molecule performs its biological functions
- design new drugs

Biological experiments of PPI site are

- time-consuming
- very expensive

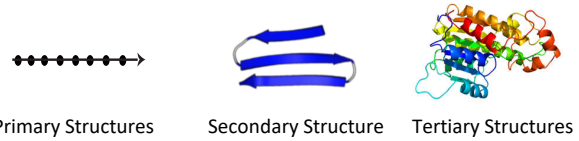
Motivations and Goal Similar shape corresponds to similar functions and behaviors

Structural comparison supports an improvement of the performance of PPI site prediction approaches:
PPI site prediction approach structure based depends on the similarity of protein architecture

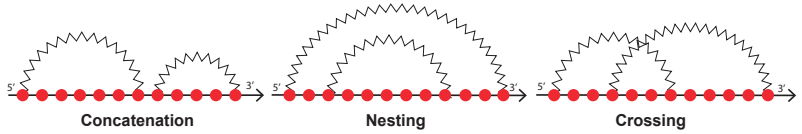
Material and Methods

Shapes can be abstracted as **secondary structure that**

- can be handled efficiently
- contains relevant biological information



We use the ASPRA distance [2] to quantify the structural differences among proteins based on three relations between arcs on a sequence



We represent the protein secondary structure representation an **arc annotated sequence**, and split each residue into as many residues as the number of bonds it forms without adding new crossings

Agglomerative Clustering allows us to classify the ASPRA distances and define new protein structural classifications

References

[1] R. Keith, et al. "Molecular biology of the cell." New York: Garland Science 32.2, 2002

[2] M. Quadrini et al. An algebraic language for RNA pseudoknots comparison. BMC bioinformatics 20.4 (2019): 1-18.

[3] M. Quadrini et al. Hierarchical Representation and Graph Convolutional Networks for the Prediction of Protein–Protein Interaction Sites. In International Conference on Machine Learning, Optimization, and Data Science (pp. 409-420). Springer, Cham

Results. A protein structural classifications based on structural differences of proteins to improve the performance of PPI site prediction approaches based on protein structure

Future work. Definition of a benchmark for PPI prediction approaches based on structural similarity among proteins